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MILITARY SPECIFICATION

INPUT DISPLAY MATRIX: 10526190

1. SCOPE

1.1 This specification covers one type of input display matrix, part of the Gun Direction Computer: M18. The matrix is used to select various input modes of the computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-F-14252	Fire Control Materiel: General Specification Governing the Design and Inspection of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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DRAWINGS

U.S. Army Munitions Command

F10526190 Input Display Matrix

PACKAGING DATA SHEET

MIL-P-14232/P10526190 Packaging of Input Display Matrix

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.1.1 Connectors.- All wiring shall be brought out to a DCM 37-P (Cannon Electric Company or equivalent) connector, which shall be attached to the equipment. Cannon female locks, No. 20418-2 (two required) shall be used to attach the connector so that the mating connector can be rigidly clamped on.

3.1.2 Hardware.- Stainless steel fasteners (bolts, screws, nuts, washers, etc.) shall be utilized throughout the construction of the equipment.

3.1.3 Solid-state devices.- The equipment design shall make full use of solid-state devices, where their use will aid in accomplishing the requirements of weight, size, reliability, power, cooling, etc. However, the use of solid-state devices shall not alter the equipment in such a manner that it will not meet the requirements of this specification

3.2 Design and construction.- The matrix shall be designed and constructed in accordance with Drawing F10526190 and drawings pertaining thereto.

3.2.1 Fire hazard.- The design and construction of the equipment shall be such that its use shall not, under any conditions of storage or operation, create a fire hazard.

3.2.2 Special tools.- The matrix shall require no special tools for assembly, dis-assembly or maintenance.

3.2.3 Mounting attitude.- The matrix shall operate satisfactorily when mounted in any position.

3.2.4 Finishing.- All external metallic surfaces shall be painted olive drab Color No. 34087 (lusterless) in accordance with FED-STD-595.

3.2.5 Electrical design and construction.

3.2.5.1 Schematic.- Schematic diagram shown on Drawing F10526190 depicts an outline which will satisfy the requirements of this specification. This is not mandatory, but is offered as a means of clarifying the requirements defined by this specification.

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3.2.5.2 Reference designations.- Reference designations shall be in accordance with MIL-STD-16. The unit numbering method described in MIL-STD-16 shall apply. The assignment of reference designations shall be made on schematic diagrams drawn in conformance with MIL-D-70327.

3.2.5.3 Electrical grounding.- Chassis ground and matrix wiring shall be kept electrically isolated with an equivalent impedance of 10 megohms or greater. Chassis ground shall return through the connector on a separate pin.

3.2.5.4 Input power.- The matrix shall be designed to operate from a power source of +15 ± 1.05 Vdc and a maximum ripple of 1 percent peak-to-peak. Current shall not exceed 1 ampere peak.

3.2.6 Size and mounting.- The matrix shall conform to the configuration shown in Drawing 10526190.

3.2.7 Weight.- The weight of the matrix shall not exceed seven pounds.

3.2.8 Manual controls.- The following manual controls shall be provided for the matrix unit.

3.2.8.1 Matrix.- The matrix shall be an eight by eight matrix consisting of sixty-four intersections or positions. Each of these sixty-four positions shall correspond to a different parameter.

3.2.8.1.1 Selection of parameter.- Each parameter or position shall be capable of being selected by the actuation of pushbutton switches provided on the matrix.

3.2.8.2 Indication of selection.- Selection of a given parameter shall result in a visual indication of such selection by the illumination of the location associated with that parameter.

3.2.8.3 Interlock.- Provisions shall be made to assure that only one parameter may be activated at a given time. The selected parameter shall remain activated until relieved by the activation of a different parameter.

3.2.8.4 Matrix pushbuttons.

3.2.8.4.1 Shape.- All pushbuttons shall have a concave actuating surface comfortable to the touch and designed to minimize any tendency for the finger to slip from the pushbutton.

3.2.8.4.2 Operating pressure.- The operating pressure required to operate each matrix pushbutton shall be 2.0 ± 1.0 pounds throughout the temperature range of from -40 to +135 degrees Fahrenheit (F).

3.2.8.4.3 Feel.- Operation of each matrix pushbutton shall result in a positive "feel" indicating that the pushbutton has been depressed sufficiently to accomplish actuation of the switch.

3.2.8.4.4 Travel.- The travel of each matrix pushbutton shall be 1/4 ± 1/32 inch.

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3.2.8.4.5 Size.- The size of the pushbutton shall conform with center-to-center spacing and minimum clearance between pushbuttons as outlined in the following paragraph.

3.2.8.4.6 Spacing.- The center-to-center spacing of pushbuttons shall be a minimum of 5/8 inch. There shall be a minimum clearance between pushbuttons of 1/8 inch.

3.2.8.4.7 Color.- The color of each matrix pushbutton shall be lusterless black, Color No. 37038, FED-STD-595.

3.2.8.5 Light intensity.- Good transmission of light shall be provided for each location to assure ease of identification of the lighted window under bright ambient sunlight conditions.

3.2.8.5.1 Spillover.- The lighting of each window location shall be such that a minimum spillover into adjacent window locations shall occur.

3.2.8.5.2 Lamps.- Lamps used for illumination shall be Part Number 8625586.

3.2.8.5.3 Lamp replacement.- It shall be possible to replace any individual illumination source in the field from the front of the panel.

3.2.8.5.4 Spare Parts.- The matrix shall be designed to provide storage for a minimum of three spare lamps and six spare diodes.

3.2.8.5.4.1 Location of spare parts.- Spare parts containers should be flush mounted to the front panel and should require no special tools for removal of parts.

3.2.8.6 Other pushbuttons.- In addition to the pushbuttons described in 3.2.9 of this specification, there shall be seven other pushbuttons (see Figure 1).

Group I - Five pushbuttons
Group II - Two pushbuttons

3.2.8.6.1 Design requirements.- The design requirements for these seven pushbuttons are identical to the design requirements for matrix pushbuttons described in 3.2.8.4 (size and shape).

3.2.8.6.2 Lettering.- All lettering on pushbuttons shall be permanent white characters (lusterless white, Color No. 37875, FED-STD-595) flush to the surface and shall be 3/16 inch high commercial Gothic type.

3.2.8.6.3 Interlock Group I (5 pushbuttons).- The pushbuttons for Group I shall be interlocked in such a way as to assure that only one pushbutton may be activated at a given time. The activated pushbutton shall remain in the actuated position to provide visual indication of its selection until relieved by the actuation of another pushbutton. The action of the interlock shall be such that as a pushbutton is being operated, the switch that was formerly activated shall return to the off position (back contact closed) prior to opening of the newly selected back contact.

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3.2.9 Switch controls.- The following switching controls shall be provided:

3.2.9.1 Matrix.- The matrix shall utilize sixteen pushbutton switches (see Figure 1).

3.2.9.1.1 Location of switches.- The sixteen switches shall be arranged in a column of eight (8) switches and a row of eight (8) switches. Each switch in the column shall be located immediately to the left of a matrix row and each switch in the row shall be located beneath a matrix column. (see Figure 1).

3.2.9.1.2 Activation of a parameter.- Activation of a parameter shall be accomplished by the actuation of the switch in the column and the switch in the row that correspond to the coordinates of the parameter location in the matrix.

3.2.9.1.3 Diodes.- Diodes shall be replaceable from the front of the panel without unsoldering leads. The ends of the diode should be encapsulated to afford broader and more rugged area for pressure contact. Diodes shall be Part No. 10526767. Leads shall not act as pressure contacts.

3.2.9.2. Contacts.

3.2.9.2.1 Sealing.- All switch contacts shall be sealed against the environmental conditions.

3.2.9.2.2 Plating.- All switch contacts shall be gold plated to withstand the conditions of MIL-STD-202, Method 101A, Test Condition B.

3.2.9.2.3 Current capacity.- All switch contacts shall be capable of carrying all values of current within the range of 0 to 500 ma, at 24 vdc continuous duty.

3.2.10 Interference control.- Generation of electrical interference by the equipment and the vulnerability of the equipment to electrical interference shall be controlled within the electrical limits of MIL-F-13926.

3.2.11 Interface connector terminations.- The interface connectors as shown on Drawing 10526190 shall be wired in accordance with Table I.

3.3 General specifications.- The matrix shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.3.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

3.3.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Handling

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3.4 Environmental service conditions.- Unless otherwise specified the matrix shall meet the following requirements at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

3.4.1 Vibration.- The matrix shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the matrix for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the matrix shall not be damaged or adversely affected and shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2.

3.4.2 Shock.- The matrix shall withstand 3 shock impulses in each of 3 mutually perpendicular directions. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2.

CONNECTOR PIN - MATRIX POSITION RELATIONSHIP

TABLE I

ACTIVATED PIN NO.	Activated location No. (See Figure 1) All activated pin voltages to be -10 VDC
1	1 2 3 4 5 6 7 or 8
2	9 10 11 12 13 14 15 or 16
3	17 18 19 20 21 22 23 or 24
4	25 26 27 28 29 30 31 or 32
5	33 34 35 36 37 38 39 or 40
6	41 42 43 44 45 46 47 or 48
7	49 50 51 52 53 54 55 or 56
8	57 58 59 60 61 62 63 or 64
9	57 49 41 33 25 17 9 or 1
10	58 50 42 34 26 18 10 or 2
11	59 51 43 35 27 19 11 or 3
12	60 52 44 36 28 20 12 or 4
13	61 53 45 37 29 21 13 or 5
14	62 54 46 38 30 22 14 or 6
15	63 55 47 39 31 23 15 or 7
16	64 56 48 40 32 24 16 or 8
17	Switch A (N.C.)
18	Switch A (N.O.)
19	Switch B (N.C.)
20	Switch B (N.O.)
21	Switch C (N.C.)
22	Switch C (N.O.)
23	Switch D (N.C.)

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ACTIVATED PIN NO.	Activated location No. (See Figure 1) All activated pin voltages to be -10 VDC
24	Switch D (N.O.)
25	Switch E (N.C.)
26	Switch E (N.O.)
27	Switch 1
28	Switch 2
30	GROUND Chassis
35	(-10 VDC) (Common Switch Supply)
36	(15 VDC) (Common Lamp Supply)
37	(15 VDC) (Common Lamp Supply)
29	
31	
32	Spares
33	
34	

3.4.3 Temperatures.

3.4.3.1 Operating.- The matrix shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2 while exposed to temperatures of -40°F and +135°F.

3.4.3.2 Storage.- The matrix shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.4.4 Humidity.

3.4.4.1 Operating.- The matrix shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.4.4.2 Non-operating.- The matrix shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.4.5 Fungus resistance.- The matrix shall show no evidence of damage and shall meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2 after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 $\pm$ 2%.

3.4.6 Salt spray.- The matrix shall show no evidence of physical damage and meet the requirements of 3.2.5.4, 3.2.8.2, 3.2.8.3, 3.2.8.4.2, 3.2.8.6.3, and 3.2.9.1.2 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4.7 Sand and dust.- The matrix, simulated in its computer mounting position, shall be capable of operation for 6 hours in an atmosphere heavily laden with sand and dust particles at an ambient temperature of 77° $\pm$ 15°F. It shall also be capable of operation after exposure for 6 hours to the sand and dust conditions at 160°F.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables II, III, IV, and V and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE II
(100 percent inspection)

	Characteristics	Requirement	Test Procedures
101.	Operating pressure (+60° to +90°F)	3.2.8.4.2	4.5.3.2.3
102.	Matrix, switch controls, interlock	3.2.8.2, 3.2.8.3, 3.2.8.6, 3.2.8.6.3 and 3.2.9.1.2	4.5.3.2.1, 4.5.3.2.2
103.	Current capacity	3.2.9.2.3	4.5.3.3
104.	Interference control	3.2.10	4.5.3.4

4.2.5 Certification provisions.- Acceptance inspection for material, design and construction in compliance with 3.1 and 3.2 shall be in accordance with 4.5.2.1 and Table IV.

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4.3 Standard sampling.

4.3.1 General specifications.- One matrix shall be selected at random by a Government representative as a standard sample from each 100 produced to determine compliance with 3.3. A visual examination shall be conducted to determine conformance with MIL-F-13926, and MIL-C-46357 as specified in 3.3.1 and 3.3.2.

4.3.2 Materials and construction.- Three units shall be selected at random by a Government representative as standard samples from each 100 produced or from each month's production, whichever occurs first. Examination of the standard samples shall be as specified in 4.5.2.2 and Table V for each characteristic listed.

4.3.3 Environmental.- Three units shall be selected at random by a Government representative as standard samples from each 50 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

TABLE III

Characteristics	Requirement	Test Procedures
301. Vibration	3.3.1	4.5.1.1
302. Shock	3.3.2	4.5.1.2
303. Humidity-Operating	3.4.4.1	4.5.1.3.2, 4.5.3.2 to 4.5.3.2.3 incl.
304. Humidity-Non-operating	3.4.4.2	4.5.1.3.1
305. Salt spray	3.3.6	4.5.1.4
306. Operating temperature	3.4.3.1	4.5.1.5, 4.5.3.2 to 4.5.3.2.3 inclusive
307. Storage temperature	3.4.3.2	4.5.1.5
308. Sand and dust	3.4.7	4.5.1.7, 4.5.3.2 to 4.5.3.2.3 incl.

4.3.4 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented inspection log will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.4.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.4.1 with the matrix rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.1.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.4.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5G's. The matrix shall be subsequently inspected for physical damage or defects in compliance with 3.4.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with requirements specified under 3.4.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.4.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dired and then subjected to the test in 4.5.1.3.2 prior to meeting the requirements specified under 3.4.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.4.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 $\pm$ 5 percent for the 24 hour period and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3.2 to 4.5.3.2.3 inclusive to determine compliance with 3.4.4.1.

4.5.1.4 Salt spray.- The matrix shall be subjected to the conditions in 3.4.7 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.3.2 to 4.5.3.2.3 inclusive for compliance with 3.4.7.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the matrix shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.4.3 shall apply. While at the operating temperatures of 3.4.3.1, the unit shall be tested in accordance with 4.5.3.2 to 4.5.3.2.3 inclusive to determine compliance with the requirements under 3.4.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.3.2 to 4.5.3.2.3 inclusive to determine compliance with 3.4.3.2.

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4.5.1.6 Fungus.- To determine compliance with 3.4.5, one of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.1.7 Sand and dust.- This test shall be conducted in accordance with the sand and dust test defined in MIL-F-13926 to determine compliance with 3.4.7. Temperature changes shall not occur at a rate exceeding 2 degrees Fahrenheit per minute to avoid thermal shock. The matrix shall be positioned in the same manner as it is positioned when functioning as a part of Computer Set, Gun Direction: M18. Operation while at 77°F shall be conducted in accordance with the tests in 4.5.3.2 to 4.5.3.2.3 inclusive.

4.5.2 Materials, design and construction.

4.5.2.1 Certification.- Certification shall include material, process or test data and results for all characteristics specified in Table IV. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor and Government approval therefore.

TABLE IV

Characteristics	Requirement
Materials	3.1
Connector	3.1.1
Hardware	3.1.2
Solid-state device	3.1.3
Fire hazard	3.2.1
Special tools	3.2.2
Electrical grounding	3.2.5.3
Indication of selection	3.2.8.2
Interlock	3.2.8.3
Sealing (contacts)	3.2.9.2.1
Plating (contacts)	3.2.9.2.2

4.5.2.2 Examination.- Measurement by Standard Measuring Equipment (SME) as well as visual and tactile examinations shall be conducted for all requirements listed in Table V (see 4.3.4).

TABLE V

Characteristics	Requirement	Examination
Mounting attitude	3.2.3	Visual, Tactile, SME
Surface finish	3.2.4	FED-STD-595, Visual
Schematic	3.2.5.1	Drawing, Visual
Reference designations	3.2.5.2	MIL-STD-16, MIL-D-70327, Visual
Size and mounting	3.2.6	Drawing, Visual, SME
Weight	3.2.7	SME

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Characteristics	Requirement	Examination
Matrix, positions	3.2.8.1	Visual
Shape, pushbuttons	3.2.8.4.1	Visual, tactile
Feel, pushbuttons	3.2.8.4.3	Tactile
Travel, pushbuttons	3.2.8.4.4	SME
Size and spacing of pushbuttons	3.2.8.4.5, 3.2.8.4.6	SME
Color of pushbuttons	3.2.8.4.7	FED-STD-595, Visual
Light intensity	3.2.8.5	Visual
Spillover	3.2.8.5.1	Visual
Lamps, part number	3.2.8.5.2	Visual
Lamp replacement	3.2.8.5.3	Tactile
Spare parts	3.2.8.5.4	Visual
Location of spare parts	3.2.8.5.4.1	Visual, Tactile
Pushbuttons, other	3.2.8.6	Visual
Size and shape, other pushbuttons (7)	3.2.8.6.1	Visual, SME
Lettering	3.2.8.6.2	FED-STD-595, Visual, SME
Switches	3.2.9.1	Visual
Location of switches	3.2.9.1.1	Visual
Diodes	3.2.9.1.3	Visual, Tactile
Interface connector terminations	3.2.11	Visual, Table & Dwg.

4.5.3 Performance.

4.5.3.1 Input power.- Unless otherwise specified herein, the matrix shall be tested with a power source as defined in 3.2.5.4.

4.5.3.2 Matrix and switch controls.- To determine compliance with 3.2.8.2 and 3.2.9.1.2 utilize the following two loads which are defined for the selected parameters shall be applied:

(a) Type A - Internal lamp load across a 15 VDC supply.
(b) Type B - External 47 ohm resistor load across a 10 VDC supply. (213 ma load for switch contacts).
(c) Type A and B loading of selected parameters shall be as follows:

(1) Apply a lamp voltage of +15 ± 1.05 VDC to pin 36 and the negative return to pin 37.
(2) Apply a -10 ± 1.0 VDC to pin 35 and a 47 ± 5 percent ohm resistor load to ground. (see 3.2.5.4).

4.5.3.2.1 Matrix.- To determine compliance with 3.2.8.2, 3.2.8.3, and 3.2.9.1.2 activation of one switch in a row and one in a column that corresponds to the coordinates of the parameters shall provide a visual indication by illuminating the matrix window of the selected parameters. A -10 ± 1.0 VDC level shall be measured across the corresponding resistor loads. All other resistor loads shall measure 0 ± 0.1 VDC.

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4.5.3.2.2 Switch controls (pushbuttons) and interlock.

4.5.3.2.2.1 Group I pushbuttons.- To determine conformance with 3.2.8.6.3 the test be conducted utilizing the five back contacts (pins 17, 19, 21, 23 and 25) which represents the Group 1 pushbuttons (see Figure 1). The readings shall be taken at the five contact pins to determine the proper open and closed position of each contact. Compliance with 3.2.8.6.3 shall be determined when depression of any one pushbutton which has not been actuated (depressed), causes the actuated (depressed) pushbutton to return to the original (non actuated) position. All back contacts shall close for at least 10 microseconds before the new back contact opens. When actuated, the contact representing the actuated pushbutton (switch) shall be open. All other contacts shall be closed. The interlock shall not permit more than one pushbutton to be actuated at any one time to determine compliance with 3.2.8.3 and 3.2.8.6.3.

4.5.3.2.2.2 Other pushbuttons.- To determine compliance with 3.2.8.6, activation of any one of seven switches (Group I and Group II) shall cause a -10 $\pm$ 1.0 volt level to appear across the 47 $\pm$ 5 percent ohm resistor load. All other resistor loads shall measure 0 $\pm$ 0.1 volts.

4.5.3.2.3 Operating pressure.- Test equipment to test for compliance with 3.2.8.4.2 shall be capable of measuring direct force of the pressure or push (in pounds), within a minimum range of zero to ten pounds. The accuracy of measurement shall be $\pm$ 2 ounces throughout the entire zone of operation. The pressure required to operate the matrix pushbuttons from their normal position shall not exceed the limit specified in 3.2.8.4.2 for the temperatures indicated.

4.5.3.3 Current capacity.- All switch contacts shall be tested under a continuous duty condition while energized by 24 Vdc and shall not breakdown or be adversely affected when subjected to values of current ranging from 0 to 500 ma to determine compliance with 3.2.9.2.3.

4.5.3.4 Interference control.- The electrical tests in MIL-F-13926 shall be utilized to determine compliance with 3.2.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10526190.

6. NOTES

6.1 Intended use.- The matrix shall have the capability of providing manual and visual coded selection of numerous parameters by actuation of pushbutton switches. This information will be used to select various input modes of a computer.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each input display matrix.

Custodian:

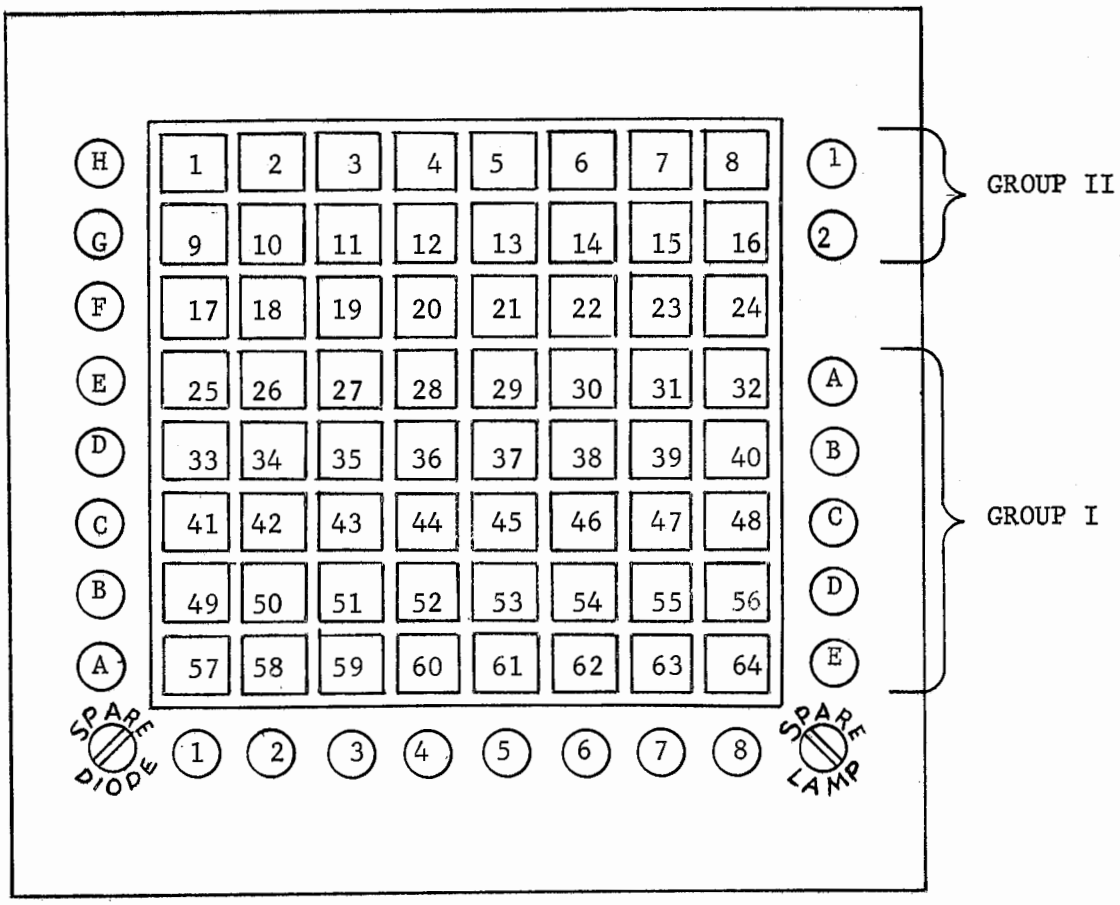
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NOTE: Numbered squares are for locational purposes only.

INPUT DISPLAY MATRIX

FIGURE 1

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